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Substrate Hydrolysis by Matrix Metalloproteinase-9*

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The catalytic clefts of all matrix metalloproteinases (MMPs) have a similar architecture, raising questions about the redundancy in substrate recognition across the protein family. In the present study, an unbiased phage display strategy was applied to define the substrate recognition profile of MMP-9. Three groups of substrates were identified, each occupying a distinct set of subsites within the catalytic pocket. The most prevalent motif contains the sequence Pro-*X*-*X*-Hy-(Ser/Thr) at P_3 through $P_{2'}$. This sequence is similar to the MMP cleavage sites within the collagens and is homologous to substrates the have been selected for other MMPs. Despite this similarity, most of the substrates identified here are selective for MMP-9 over MMP-7 and MMP-13. This observation indicates that substrate selectivity is conferred by key subsite interactions at positions other than P_3 and $P_{1'}$. This study shows that MMP-9 has a unique preference for Arg at both P_2 and P_1, and a preference for Ser/Thr at $P_{2'}$. Substrates containing the consensus MMP-9 recognition motif were used to query the protein data bases. A surprisingly limited list of putative physiologic substrates was identified. The functional implications of these proteins lead to testable hypotheses regarding physiologic substrates for MMP-9.

Matrix metalloproteinase-9 (MMP-9)[1] is a member of the matrixin family of metallo-endopeptidases (1–3). MMP-9 is historically referred to as gelatinase B because of its ability to cleave gelatin, a denatured form of collagen, *in vitro*. Along with MMP-2, MMP-9 differs from other MMPs because it contains three fibronectin type II repeats that have high binding affinity for collagen. These repeats are thought to mediate the binding of MMP-2 and -9 to collagen (1, 2). This binding interaction brings the catalytic pocket of the MMP in proximity to collagen, thereby enhancing its rate of hydrolysis. Despite these well characterized biochemical interactions, it is now clear that MMP-9 is also able to cleave a number of other proteins and may have a rather wide range of physiologic substrates (4–8).

Much of our understanding of the biological function of MMP-9 comes from the study of mice lacking this gene. For example, MMP-9-deficient mice have impaired ossification of the skeletal growth plate, a defect that has been partially attributed to poor vascularization of developing bone (9). Studies on these mice also show that MMP-9 is essential for the recruitment of osteoclasts into developing bones (10). Other work indicates that MMP-9-deficient mice are resistant to dermal blistering in a bullous pemphigoid model, an effect that has been attributed to the inability of these mice to cleave the SERPIN α1-proteinase inhibitor (5). Finally, recent studies in the RIP1-Tag2 transgenic mouse model of multistage carcinogenesis indicate that MMP-9 is part of the angiogenic "switch" that is essential for tumor growth (11, 12). Other reports suggests that MMP-9 may play a role in inflammation in the nervous system. MMP-9 is elevated in encephelomyelitis (7, 8), in the cerebrospinal fluid of patients with multiple sclerosis (13), and in patients with AIDS-related dementia (14).

A first approximation of the substrate recognition specificity of MMP-9 has been gleaned from alignments of its cleavage site within a number of different proteins (4, 8, 9, 15–17). Nevertheless, a detailed understanding of subsite preferences for MMP-9 is lacking. Such an analysis is particularly important because the catalytic cleft of MMP-9 is closely related to other MMPs, raising questions about the distinction among substrates for these proteases. These issues are particularly important because many of the current pharmacologic antagonists of the MMPs have overlapping inhibition profiles. Furthermore, in the post-genomic era, where the sequences of all proteins will soon be available, information on substrate recognition could help identify important physiologic substrates.

With these ideas in mind, we applied an unbiased approach to define the substrate recognition preference of MMP-9. MMP-9 was used to cleave substrates within a vastly complex phage display library of random hexamers. Substrates within this library can be cleaved at any position within the hexamer, allowing information on the substrate specificity on both sides of the scissile bond to be obtained. Three families of substrates were identified. The largest group contains a Pro-*X*-*X*-Hy-(Ser/Thr) motif (*X* is any residue, and Hy is a hydrophobic residue) that occupies positions P_3 through $P_{2'}$. This general motif is cleaved by a number of MMPs and is presumed to represent a collagen-like substrate (18–23). Nevertheless, substrates within this family that were selected show considerable selec-

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[1] The abbreviations used are: MMP, matrix metalloproteinase; PCR, polymerase chain reaction; BSA, bovine serum albumin; ELISA, enzyme-linked immunosorbent assay; TBS-T, Tris-buffered saline with Tween 20; MALDI-TOF, matrix-assisted laser desorption ionization/time of flight; APMA, p-Aminophenylmercuric acetate.

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tivity for MMP-9. The second group of substrates are defined by a Gly-Leu-(Lys/Arg) motif at positions P_1 through $P_{2'}$. Members of the third family of substrates are unique in that they contain Arg residues at both P_1 and P_2. Altogether, these findings reveal multiple modes of substrate recognition by MMP-9 and provide important insights into the hydrolysis of physiologic substrates that may be important in biology and pathology.

EXPERIMENTAL PROCEDURES

Purified forms of full-length MMP-7, MMP-13, and TIMP-2 were purchased from Chemicon (Temecula, CA). Ilomastat (GM6001) was purchased from AMS Scientific (Concord, CA). Restriction enzymes were from Roche Biosciences or New England Biolabs. Oligonucleotides were synthesized by Integrated DNA Technologies, Inc. Tissue culture media and reagents were from Irvine Scientific (Irvine, CA). All other reagents, chemicals, and plasticware were from Sigma or Fisher.

Construction of Substrate Phage Display Library—Substrate phage libraries were generated using a modified version of the fUSE5 phagemid (24–26). A FLAG epitope was engineered at the NH_2 terminus of the geneIII protein by annealing oligonucleotides 5′-CCGGGTTTGTC-GTCGTCGTCTTTGTAGTCGGTAC-3′ and 5′-CGACTACAAAGACGA-CGACGACAAAC-3′ and ligating them into fUSE5 at the *Kpn*I and *Xba*I restriction sites. The random hexamers were generated by PCR extension of the template oligonucleotide 5′-GGGGAGGCCGACGTGGCC-GTCATCAGGCGGCTCAGGC$(NNK)_6$ACGG CCTCTGGGGCCGAAAC-3′, where N is any nucleotide and K is either G or T. The template oligonucleotide also encodes an SGGSG linker positioned in between the FLAG epitope and the random hexamer. A primer oligonucleotide 5′-AATTTCTAGTTTCGGCCCCAGAGGC-3′ and the template oligonucleotide were mixed and heated at 65 °C for 2 min. The heating block was switched off and allowed to cool passively to 40 °C to allow annealing of the extension oligonucleotide to the template oligonucleotide. Elongation of the template oligonucleotide was performed using Sequenase (United States Biochemical Corp.) (27). The final cDNA product was precipitated with ethanol, re-suspended in water, and digested with *Sfi*I. The DNA insert and fUSE5 were mixed and ligated at a 5:1 molar ratio and electroporated into *Escherichia coli* MC1061(F−). Several phage were selected for sequencing to confirm randomness in the insert sequences and the correct reading frame.

Expression and Purification of MMP-9—The cDNA encoding the catalytic domain of MMP-9 (28, 29) was generated by PCR and cloned into pCDNA3 (Invitrogen). The expression vector was used to transfect HEK 293 cells by electroporation, and individual antibiotic-resistant colonies were isolated with cloning rings, expanded, and then screened by reverse transcription-PCR for MMP-9 mRNA. Expression of the proteinaceous catalytic domain was determined by zymography using conditioned medium from the transfected cells. Cells expressing the MMP-9 catalytic domain were grown in 150-mm dishes and conditioned in Dulbecco's modified Eagle's medium with 10% fetal bovine serum supplemented with G418 (200 μg/ml) for 5 days. The catalytic domain of MMP-9 was purified from the conditioned medium by gelatin-Sepharose chromatography as described previously (28, 29). Fractions containing MMP-9 were pooled and dialyzed against 50 mM Tris, pH 7.8, 10 mM NaCl. The dialyzed material was then purified further using a 1-ml Hi-Trap Source Q ion exchange column (Amersham Pharmacia Biotech). Fractions containing MMP-9 were pooled and dialyzed against 50 mM Tris, pH 7.5, 100 mM NaCl, 10 mM $CaCl_2$ and then concentrated in a dialysis bag using Aquacide II (Calbiochem). The final concentration of purified material was typically 1.5–3 mg/ml.

Activation and Active Site Titration of Proteases—The purified catalytic domain of MMP-9 was activated by incubation with 2 mM APMA for 16 h at room temperature (29). Activation was monitored by altered migration of the protein on SDS-polyacrylamide gel electrophoresis and zymography. Pro-MMP-13 was activated with APMA for 3 h at room temperature, and MMP-7 required no activation.

The level of active protease was always quantified by active site titration prior to kinetic studies. The active site of MMP-9 was titrated using the hydroxamate inhibitor Ilomastat (30). Briefly, 25 nM MMP-9 was incubated with a range of Ilomastat. Inhibition of MMP-9 by Ilomastat was then measured by determining the rate of cleavage of three concentrations of the peptide, m1A11, using the fluorescamine incorporation assay (21, 31). The active sites of MMP-13 and MMP-7 were titrated with human TIMP-2. Briefly, 5–15 nM of each protease were pre-incubated with a range of TIMP-2 for 5 h at room temperature. Residual MMP-13 activity was measured by cleavage of MCA-Pro-Cha-Gly-Nva-His-Ala-Dpa-NH_2 (Calbiochem) as substrate, and residual MMP-7 activity was measured by cleavage of MCA-Pro-Leu-Gly-Leu-Dpa-Ala-Arg-NH_2 (Calbiochem) with monitoring at λ_{ex} 320 nm and λ_{em} 405 nm.

Phage Selection of MMP-9 Substrates—The substrate phage library (2×10^{10} phage) was incubated with 2.5 μg/ml MMP-9 in 50 mM Tris, pH 7.4, 100 mM NaCl, 10 mM $CaCl_2$, 0.05% Brij-35, and 0.05% BSA for 1 h at 37 °C. A control selection was performed without protease. The cleaved phage were separated from the non-cleaved phage by immunodepletion. 100 μg of anti-FLAG monoclonal antibody (Sigma) was added to the phage samples and then incubated for 18 h with rocking at 4 °C. The phage-antibody complexes were precipitated by the addition of 100 μl of Pansorbin (Calbiochem). The cleaved phage remaining in the supernatant were amplified using K91 *E. coli* as described previously (32–34) and were then used for subsequent rounds of substrate selection.

Substrate Phage ELISA—Hydrolysis of individual phage substrates was measured using a modified ELISA. 96-well microtiter plates were coated with anti-M13 antibody (Amersham Pharmacia Biotech, 2.5 μg/ml), in phosphate-buffered saline, overnight at 4 °C. After coating, the wells were blocked for 1 h at room temperature in TBS-T (50 mM Tris, pH 7.8, 150 mM NaCl, 0.2% Tween 20) containing 10 mg/ml BSA. After blocking, 150 μl of supernatant from an overnight phage culture was added to each well and incubated for 2 h at 4 °C to allow for phage capture. Unbound phage were removed with three washes of ice-cold TBS-T. To asses hydrolysis, MMP-9 (2.5 μg/ml) was added to the appropriate wells in Incubation Buffer (50 mM Tris, pH 7.5, 100 mM NaCl, 10 mM $CaCl_2$, 0.05% BSA, 0.05% Brij-35, 50 μM $ZnCl_2$) for 2 h at 37 °C. Control wells lacked protease. The protease solution was removed, and the wells were washed four times with ice-cold TBS-T. To measure hydrolysis of the peptides on phage by MMP-9, anti-FLAG polyclonal antibody (1.8 μg/ml in TBS-T with 1 mg/ml BSA) was added to each well and the plates were incubated at 4 °C for 1 h. Binding of anti-FLAG antibody to FLAG epitope was measured with a horseradish peroxidase-conjugated goat anti-rabbit IgG antibody (Bio-Rad) followed by detection at 490 nm. The extent of hydrolysis, taken as a measure of substrate hydrolysis, was calculated by the ratio of the optical density at 490 nm of the protease-treated samples *versus* samples lacking protease.

Determination of Scissile Bonds—The cleavage site for MMP-9 within peptide substrates was determined using MALDI-TOF mass spectrometry. MMP-9 (25 nM) was incubated with 100 μM amounts of each peptide, independently, in 50 mM Tris, pH 7.5, 100 mM NaCl, 10 mM $CaCl_2$ for 2 h at 37 °C. The mass of the cleavage products was determined using a Voyager DE-RP MALDI-TOF mass spectrometer (PerSeptive Biosystems, Framingham, MA). Following hydrolysis, the peptide samples were prepared according to methods described previously (35–38). In all cases, no other peaks that corresponded to other potential cleavage products were observed.

Kinetic Measurements of Peptide Hydrolysis—The kinetic parameters of substrate hydrolysis were measured using a fluorescamine incorporation assay that has been described previously (21, 31, 34, 39). Briefly, MMP-9, MMP-7, or MMP-13 were incubated with individual peptide substrates at concentrations ranging from 100 to 800 μM in 50 mM Tris, pH 7.5, 100 mM NaCl, 10 mM $CaCl_2$, 50 μM $ZnCl_2$. At selected time points, the reactions were stopped by the addition of 1,10-phenanthroline. Peptide hydrolysis was determined by the addition of fluorescamine followed by detection at λ_{ex} 355 nm and λ_{em} 460 nm. The data were transformed to double-reciprocal plots (1/[S] *versus* $1/v_i$) to determine K_m and k_{cat} (21, 31, 34, 39). Similar results were obtained using different batches of protease. For some substrates, K_m and k_{cat} could not be determined individually, but the specificity constant, k_{cat}/K_m, was derived by the equation: $k_{cat}/K_m = v_i/(E_0)(S_0)$ (22).

Computational Molecular Modeling—The terminology established by Schechter and Berger is used to describe the subsites in the protease active site and the correlating positions in substrates (40). This method describes the former as a set of subsites designated as S (S_1, S_2, etc.) and the latter as P (P_1, P_2, etc.). The primed sites are to the carboxyl-terminal and unprimed sites are to the amino-terminal side of the scissile bond. The x-ray structure of MMP-9 has not been reported to date. We have previously reported a computational three-dimensional model for the catalytic domain of MMP-9, which was generated by homology modeling taking advantage of sequence alignment and conservation of secondary structural elements (41–43). This model shows high sequence and three-dimensional homology to the structure of MMP-2 (42), and it indicates an extended cleft for binding of substrate that can accommodate at least six amino acids from the substrate. We considered molecular models for representative members of each class of substrates (peptide A10 and peptide C11), but we concluded that there is not sufficient sequence information for the group II substrates

to draw meaningful conclusions. Therefore, the modeling work was concentrated on groups I and III. The molecular models of the sequences flanking the sites of MMP-9 cleavage in clones A10 and C11 (Ac-SGPLFYSVTA-NH_2 and Ac-SGRRLIHHTA-NH_2, respectively) were constructed using the SYBYL molecular modeling program and were docked into the active site of MMP-9 (42). Each enzyme-substrate complex contained the hydrolytic water molecule positioned in the active site near the Glu-402 side chain, the general base that promotes it for the hydrolytic reaction. The enzyme-substrate complexes were energy-minimized using the AMBER 5.0 software package (44, 45). The protocol for the energy minimization was described previously (42).

RESULTS

Construction of the Phage Library—We created a system for displaying random hexamer substrates on the surface of phage. The fUSE5 polyvalent phage display vector (24–26) was modified to express random hexamers at the amino terminus of the geneIII protein, and an octapeptide FLAG epitope at the amino-terminal end of the random hexamers. The library comprises 2.4×10^8 independent transformants, giving a 75% confidence that each of the 6.4×10^7 possible random hexamer sequences are represented in the library. Sequencing of phage confirmed the randomness of the hexamer insert. Under the selection conditions, greater than 95% of phage could be immunodepleted using an anti-FLAG antibody (data not shown).

Selection of Peptide Substrates for MMP-9—Optimal substrates were selected by exposing the phage library to a recombinant form of the catalytic domain of MMP-9 expressed in HEK 293 cells (29). The recombinant catalytic domain of MMP-9 was purified on gelatin-Sepharose, followed by ion exchange chromatography (29). The protease was activated with 2 mM APMA for 16 h at room temperature, and the active site was titrated with the hydroxamate inhibitor Ilomastat (29, 30). Phage selections were performed with 2.5 μg/ml (56 nM) active MMP-9. Following three rounds of exposure to MMP-9, individual phage clones were selected for sequencing. An alignment of the motifs revealed three groups of structurally distinct substrates (Table I). Substrates in group I contain the motif with sequence Pro-*X*-*X*-Hy-(Ser/Thr). Further analysis shows that, within this larger motif, Arg is favored at P_2 and Ser/Thr is favored at P_1 (consensus motif noted in Table I). Substrates in group II contain a Gly-Leu-(Lys/Arg) motif. Interestingly, both groups I and II are related to substrates that have been described previously for other MMPs and are somewhat related to sites of cleavage in collagen (21–23). Group III substrates, however, appear to represent a novel recognition sequence as it contains the Arg-Arg-Hy-Leu (group IIIA) and Arg-X-Leu (group IIIB) motifs. The two subclasses that comprise group III have not been described previously as a substrate motif for MMP-9.

The ability of MMP-9 to hydrolyze each of the phage clones was assessed in a semiquantitative manner using a modified ELISA. Individual phage clones were captured into microtiter plates using anti-M13 antibody. Captured clones were exposed to MMP-9 (2.5 μg/ml). A polyclonal anti-FLAG antibody was used to measure the liberation of the FLAG epitope by hydrolysis of the substrate with MMP-9. Results are expressed as the extent of hydrolysis relative to untreated phage clones (Table I). In general, the group I substrates were hydrolyzed most efficiently. However, a number of the substrates within groups II and III were cleaved to the same degree as group I substrates.

Identification of Scissile Bonds within MMP-9 Substrates—To identify the position of the scissile bonds, 10 peptides, representing each of the three groups of substrates, were designed and synthesized. Following exposure to MMP-9, the scissile bond was determined by analyzing the cleavage products by MALDI-TOF analysis (Table II). This analysis revealed

TABLE I

Sequence alignment and grouping of phage selected by MMP-9

Phage from the third round MMP-9 selection were selected for their ability to be cleaved by MMP-9 and sequenced. The substrates are separated into three groups based on distinguishable motifs. Residues that define each group are in bold. The ability of MMP-9 to cleave the substrates on phage was measured in a modified ELISA format. A rank order of preference within each group is demonstrated as a function of phage cleaved by MMP-9 relative to a non-treated control. This is expressed as percentage of hydrolysis. The results of two independent experiments are shown. Hy, hydrophobic amino acids.

Group	Peptide sequence									Clone	% Hydrolysis	
											1	2
I			**P**	L	F	***Y***	**S**	V		A10	100	100
	K	I	**P**	**R**	**T**	***L***	**T**			A11	100	100
			P	L	R	***L***	S	W		A12	100	100
			P	**R**	A	***V***	S	T		A6	100	99
	K	G	**P**	**R**	Q	***I***	T			C15	100	98
			P	**R**	P	***L***	S	G		D12	82	94
		W	**P**	L	G	***L***	A			D35	89	91
	F	R	**P**	**R**	**S**	***I***	T			D36	87	91
	R	L	**P**	V	G	***L***	T			D20	89	90
	R	S	**P**	K	**S**	***L***	T			D21	88	90
			P	V	W	***L***	A	A		D22	85	90
	I	H	**P**	S	**S**	***L***	T	A		D4	87	89
			P	A	**S**	***F***	T	S		D29	87	86
	G	Q	**P**	H	Y	***L***	T			B1	79.6	85
	M	K	**P**	A	**S**	***W***	T			B2	77.5	77
	T	H	**P**	Y	**T**	***M***	T			B10	74.1	82
		T	**P**	A	Y	***M***	L	T		A13	78.9	72
			P	L	Y	***L***	T			D2	79	82
			P	G	L	***I***	G	T		D1	75	85
			P	**R**	**S**	***I***	**S**	N		D24	82	82
	R	L	**P**	A	**S**	***Y***	**T**			D30	82	60
	N	P	**P**	**R**	Y	***L***	**T**			D5	70	67
			P	K	**T**	Q	I	S		D8	73	65
	S	V	**P**	**R**	H	***F***	**T**			D17	70	52
	L	L	**P**	A	W	***L***	**T**			D3	56	48
	T	H	**P**	Y	**T**	***M***	**T**			D14	50	47
	L	R	**P**	A	K	S	**T**			A18	28.9	30
	S	G	**P**	S	**T**	S	**T**			A9	13.1	5.8
Consensus			**P**	**R**	**S/T**	**Hy**	**S/T**					
II			G	S	**G**	**L**	**K**	A		A7	79.8	76
			A	M	**G**	**L**	**K**	S		B9	73.3	86
			K	V	**G**	**L**	**R**	T		B8	13.3	12
IIIA			G	**R**	**R**	L	**I**	H	H	C11	94.9	74
		H	P	**R**	**R**	S	**I**	T		C1	13.9	29
			G	**R**	**R**	L	**L**	S	R	A16	20.2	11
		A	L	**R**	**R**	L	E	T		C13	8.3	24
		F	Y	**K**	**R**	V	**L**	T		B6	11.7	11.1
			F	**R**	**R**	I	C	V		C7	17.9	0
	V	F	F	**R**	**R**	Q	T	A		C8	8.9	0
IIIB	G	L	A	**R**	N	**I**	T	A		B3	66	70
	F	G	S	**R**	Y	**L**	T	A		A19	45	50
		Q	D	**R**	Y	**L**	N	T		C6	40	27

that each substrate contains a hydrophobic residue at the $P_{1'}$ position. The substrates in group I contain Pro at the P_3 position. The single peptide from group II contained a Gly at P_1, Leu at $P_{1'}$, and a Lys at $P_{2'}$. The motifs of groups I and II, and locations of their scissile bonds, are similar to MMP cleavage sites in collagen and gelatin (21–23). Interestingly, however, the peptides from group III were all cleaved after an Arg residue, and in several cases an Arg-Arg occupied the P_2 and P_1 positions.

Kinetic Characterization of Substrate Hydrolysis by MMP-9—The Michaelis constant (K_m) and first-order rate constant of substrate peptide turnover (k_{cat}) were measured by incubating a range of each peptide with MMP-9. Peptide hydrolysis was measured by incorporation of fluorescamine onto newly formed amino termini as previously described (31, 33, 34). From these measurements, k_{cat} and K_m were derived for each peptide using double-reciprocal plots of 1/[S] *versus* $1/v_i$. Results from this

TABLE II

Kinetic analysis of MMP-9 peptide substrates based on phage clones

Each peptide was incubated with protease and hydrolysis was detected as described under "Experimental Procedures." The peptides are named according to their phage clone designation. All peptides were synthesized with acetylated NH_2 termini and amidated COOH-termini. A11m1 and A11m2 are analogues of peptide A11. Changes are indicated in bold. Scissile bonds are designated with ↓. Amino acids to the right of the ↓ are primed residues (. . .$P_{1'}$, $P_{2'}$, etc.), and amino acids to the left are non-primed residues (. . .P_2, P_1) (40). **, measurements were not made; *, not detected. Standard deviations of triplicate experiments ranged from 11% to 22%.

Peptide	Sequence	MMP-9			MMP-13	MMP-7
		k_{cat}/K_m	k_{cat}	K_m	k_{cat}/K_m	k_{cat}/K_m
		$M^{-1}\ s^{-1}$	s^{-1}	mM	$M^{-1}\ s^{-1}$	$M^{-1}\ s^{-1}$
C15	SGKGPRQ↓ITA	188,000	703	3.7	15,400	13,800
A11	SGKIPRT↓LTA	67,500	86.4	1.3	23,500	7,000
A6	SGPRA↓VSTTA	61,000	178	2.9	12,400	1,300
A11m1	SGKIPR**R**↓LTA	160,000	224	1.4	15,200	24,000
A11m2	SGKIPRT↓**A**TA	2000	9.4	4.7	**	**
A10	SGPLF↓YSVTA	25,600	20	0.5	**	**
B1	SGQPHY↓LTTA	8800	9	1	**	**
A18	SGLRPAK↓STA	5800	78	13.5	**	**
A7	SG↓LKALMITA	12,400	30	2.4	12,300	41,000
A19	SGFGSRY↓LTA	13,200	11	0.8	*	1,000
A16	SGRR↓LLSRTA	1200	**	**	**	**
C11	SGRR↓LIHHTA	*	**	**	**	**

analysis are shown in Table II. Among the peptides, the k_{cat} values ranged from 9 s^{-1} to 703 s^{-1}, and K_m values were generally in the high micromolar to low millimolar range. Overall, peptides from group I appeared to represent the best substrates, with peptide C15 having a k_{cat}/K_m ratio of 180,000 M^{-1} s^{-1}, the highest of any of the substrates. Peptide A11, which closely matches the consensus recognition motif of group I, also exhibits a high k_{cat}/K_m ratio (67,500 M^{-1} s^{-1}) and considerable selectivity for MMP-9.

Interestingly, the three peptides cleaved most efficiently by MMP-9 were from group I and contained Arg at the P_2 position. This led us to hypothesize that group I and III substrates might represent a larger set of substrates whose relationship to one another is not entirely evident from the sequences of only 100 clones. To test this idea, we synthesized a mutant peptide (A11m1) that contains a Thr → Arg substitution at P_1 within the context of the sequence of the A11 peptide. Hence, the mutant peptide contained elements of both group I (Pro at P_3 and Leu at $P_{1'}$) and group III (Arg at P_2 and P_1). Peptide A11m1 had a k_{cat}/K_m twice that of the parent peptide, an effect resulting primarily from an increase in k_{cat}. This finding suggests that the substitution of Arg at P_1 lowers the transition state energy of the protease-substrate interaction. This observation also indicates that, within the Pro-*X*-*X*-Hy-(Ser/Thr) motif, Arg residues at P_2 and P_1 are favored for MMP-9, and that the substrates from groups I and III may have a related binding mechanism.

In almost every case, the MMP-9 substrates selected from the phage library contain a hydrophobic residue at $P_{1'}$, a finding that is entirely consistent with the fact that MMPs are known to have a deep hydrophobic $S_{1'}$ pocket (1, 2, 46). To assess the contribution of this hydrophobic residues to substrate binding and to substrate turnover, we synthesized another mutant peptide (A11m2) based on the sequence of peptide A11, but containing Ala, rather than Leu, at $P_{1'}$ (Ac-SGKIPRT↓**A**TA-NH_2). This substitution had deleterious effects on both k_{cat} and K_m, and reduced the k_{cat}/K_m ratio nearly 30-fold. These results are consistent with the idea that the $S_{1'}$ subsite of MMP-9 coordinates substrate binding and also influences the rate of hydrolysis. Importantly, however, even this mutant peptide had a measurable k_{cat}/K_m ratio (2000 M^{-1} s^{-1}), indicating that efficient hydrolysis can be enacted by MMP-9 if the rest of the substrate sequence is optimal.

Kinetic Characterization of Substrate Hydrolysis by MMP-7 and MMP-13—Since many of the selected substrates contained a motif similar to that described for other MMPs (P-*X*-*X*-Hy), we measured the degree to which MMP-7 and MMP-13 could cleave these MMP-9 substrates. These two MMPs were used for comparison because substrates for both proteases have been selected using a similar phage display approach. Interestingly, most of the substrates we tested were cleaved more efficiently by MMP-9. The k_{cat}/K_m ratios ranged from 2.6- to 47-fold higher for MMP-9 than for either MMP-7 or MMP-13 (Table II). Only peptide A7 deviated from this trend. It is also worth noting that, although the substrates in group IIIA (Arg-Arg) were not cleaved rapidly by MMP-9, we were unable to detect any hydrolysis of these peptides by either MMP-7 or -13.

Modeling Substrate Interactions with MMP-9—Molecular modeling studies were conducted to help visualize how substrates might dock into the enzymatic cleft of MMP-9. Energy-minimized models of MMP-9 were constructed according to procedures described under "Experimental Procedures," and then docked with peptides A10 (Ac-SGPLFYSVTA-NH_2) representing group I and C11 (Ac-SGRRLIHHTA-NH_2) representing group III.

A primary feature of the group I substrates is a Pro residue at the P_3 position. The unique conformational features of Pro introduce the appropriate "bend" needed for the optimal substrate binding at this site. This Pro residue, occupying the S_3 subsite of MMP-9, is illustrated in Fig. 1*A* (*white arrow*). The group I substrates also contain an invariant hydrophobic residue at the $P_{1'}$ position. This residue protrudes into the deep $S_{1'}$ pocket of the protease, as depicted by the *orange arrows*.

Amino acids with long basic side chains at P_2 and P_1, such as Arg, are the defining features of the substrates in group III. Although the presence of these residues at P_2 and P_1 is somewhat surprising, the energy-minimized models support the observation that these residues bind favorably. An Arg at P_2 is likely to interact with the backbone carbonyl moieties of His-405, Gly-408, and the side chain of Asp-410, all of which contribute to the S_2 subsite within MMP-9 (*white arrow* in Fig. 1*B*). These electrostatic interactions are predicted to contribute to favorable binding of this class of substrates. Many of the group III substrates also contain an Arg residue at P_1. The favorable interaction of Arg into the S_1 subsite can be explained by the somewhat unusual nature of this subsite. It is essentially a hydrophobic binding surface that would be predicted to accommodate the hydrophobic side chains of amino acids such as Ala, Phe, and Tyr. However, the docking studies of peptides with Arg at P_2 show that the hydrophobic surface of the S_2 subsite could also bind to the extended methylene group in the

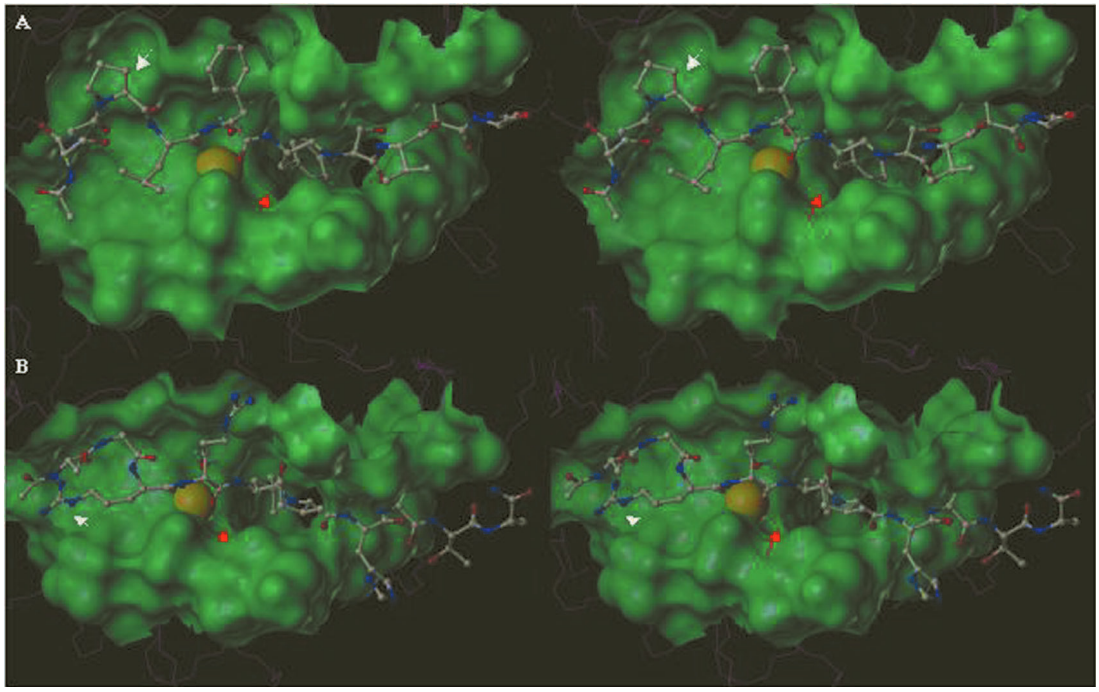


FIG. 1. **Three-dimensional models of distinct binding modes between substrate and MMP-9.** To ascertain whether peptides from different substrate families exhibit different modes of binding in the active site of MMP-9, models with two representative peptides binding to MMP-9 were constructed. The stereo view represents the energy-minimized complexes of peptide A10 (SGPLFYSVTA, *panel A*) and peptide C11 (SGRRLLSRTA, *panel B*) in the active site of the MMP-9 catalytic domain. The active site of MMP-9 is shown as a *green* Conolly surface, and the peptide atoms are colored according to type (carbon, *white*; oxygen, *red*; nitrogen, *blue*). The catalytic zinc ion is represented as a *yellow sphere*, and the backbone of MMP-9 outside the active site is shown in *magenta*. The *orange arrows* in both panels show the deep hydrophobic pocket at the $S_{1'}$ subsite. In *panel A*, the *white arrow* shows the Pro of substrate A10 penetrating well into the S_3 pocket. In contrast, a Gly from substrate C11 occupies this subsite (*panel B*).

side chain of Arg. In addition, the hydrophobic channel of the S_2 subsite contains the backbone carbonyl of Pro-180, which is likely to engage in electrostatic interaction with the basic side chain of Arg, stabilizing the interaction.

DISCUSSION

Because of their association with a number of diseases, the MMPs have received considerable attention as drug targets (15–17). Much of the effort in this area has focused on the design of small molecule antagonists with two primary features; 1) a hydroxamate moiety that binds to the proteases catalytic zinc, and 2) a rather large hydrophobic moiety that fits into the deep $S_{1'}$ pocket present in all MMPs (30). This synthetic strategy has focused structure-activity studies to essentially two positions within the catalytic pocket. An understanding of the interactions between the substrate and other key subsites within the catalytic pocket is lacking.

We have identified three families of peptide substrates for MMP-9 that each appear to interact differently with the catalytic cleft of the protease. Substrates in group I are cleaved most efficiently by MMP-9. These peptides all contain a Pro at P_3 and a large hydrophobic residue at $P_{1'}$. In this respect, the group I substrates are similar to collagen-like sequences that are known to be cleaved by the MMPs (19, 20). A prior analysis of a small series of synthetic peptides based on collagen, and containing a similar Pro-*X*-*X*-Hy core, showed k_{cat}/K_m ratios ranging from 340 to 1000 $mM^{-1}\ h^{-1}$. These values are ~100 fold lower than those exhibited by the best peptides reported here. Consequently, the added diversity afforded by phage libraries allowed us to identify better substrates and a wider range of substrates. Interestingly, many different MMPs recognize the Pro-*X*-*X*-Hy core sequence. Can subsite interactions at positions other than P_3 and $P_{1'}$ generate selectivity within this family of sequences?

Apparently, individual MMPs do exhibit a great deal of selectivity for peptides containing the P-*X*-*X*-Hy core sequence. For example, most of the MMP-9 substrates within group I are selective for MMP-9 over MMP-7 and MMP-13. Some of the k_{cat}/K_m ratios for these peptides are up to 47-fold higher for MMP-9 than for the other MMPs tested. These findings suggest that substrate selectivity can be conferred by subsite interactions outside of the dominant P_3 and $P_{1'}$ subsites that are common among MMP substrates.

Since phage substrate selections have been performed for MMP-3, -7, -9, and -13, an analysis of the frequency by which individual residues occupy distinct subsites can be used as a first test of this idea. This comparison reveals considerable distinction in the residues that occupy the P_2, P_1, and $P_{2'}$ subsites. Nearly one-third of all group I substrates for MMP-9 contain Arg at P_2. Although Arg can also be found at P_2 in peptide substrates for MMP-13, its frequency is much lower than in the MMP-9 substrates we selected (47). Furthermore, Arg is rarely, if ever, found at P_2 in MMP-3 or -7 peptide substrates (32). Ser or Thr most frequently occupies the P_1 position of the MMP-9 substrates. However, a Gly residue is preferred at this position by MMP-13, and Asp or Glu are preferred by MMP-3 and -7 (32, 47). Significant differences are also observed at $P_{2'}$. In the group I MMP-9 substrates, 23 of 28 substrates have Ser or Thr at $P_{2'}$, but neither residue is prevalent at this subsite in the substrates selected for the other MMPs (32, 47). These observations support the contention that subsite interactions other than P_3 and $P_{1'}$ have a significant impact on substrate selectivity among the MMP family.

Here we observed two additional families of substrates for MMP-9 that are distinct from the P-*X*-*X*-Hy family. Group II substrates were selected least frequently and contain only three members, making it difficult to identify subsite prefer-

TABLE III

A comparison of the MMP-9 substrate consensus with cleavage sites in known protein substrates

The group I MMP-9 consensus motif is compared to the cleavage sites in proteins known to be cleaved by MMP-9. The protein substrates are divided into two groups: collagen and non-collagen. The residues in the protein substrates that match the MMP-9 consensus are shown in bold. Scissile bonds are identified by ↓.

	P	R	S/T	↓	Hy	S/T
MMP-9 consensus	**P**	R	S/T	↓	**Hy**	S/T
collagen substrates						
α1(I) collagen	**P**	Q	G	↓	**V**	R
	P	S	G	↓	**L**	HyP
	P	A	G	↓	**V**	Q
CB3 α1(IV) collagen	**P**	S	G	↓	R	D
	P	P	G	↓	**I**	V
α1(V) collagen	**P**	P	G	↓	**V**	V
α2(V) collagen	**P**	G	G	↓	**V**	V
α1(XI) collagen	**P**	P	G	↓	**L**	R
Non-collagen substrates						
Aggrecan	**P**	E	N	↓	**F**	F
Tissue factor pathway inhibitor	**P**	L	K	↓	**L**	M
Galectin 3	**P**	G	A	↓	**Y**	H
Cartilage link protein	A	I	H	↓	**I**	Q
Myelin basic protein	H	F	F	↓	K	N
	G	L	**S**	↓	**L**	**S**
	A	S	D	↓	**Y**	K

TABLE IV

Identification of potential MMP-9 substrates

The group I MMP-9 consensus motif was used to query the SWISSPROT data base to identify potential physiological substrates of MMP-9. The search was limited to proteins that contained these potential MMP-9 cleavage sites. The putative protein substrates and potential cleavage sites are shown.

MMP-9 consensus:	P	R	S/T	Hy	S/T	Potential protein substrate (residues)
	P	R	T	I	T	Kallikrein 14 (180–184)
	P	R	T	I	S	Ladinin 1 (371–375)
	P	R	T	V	T	Endoglin (231–235)
	P	R	T	I	S	Endothilin receptor (82–86)
	P	R	S	L	T	Laminin α3 chain (1568–1572)
	P	R	S	L	S	Phosphate regulating neutral endopeptidase (566–570)
	P	R	T	I	S	ADAM 2 (299–302)
	P	R	S	L	T	Desmoglein 3 (579–583)
	P	R	S	I	T	Integrin β_5 (57–61)

ences outside of the Gly-Leu-(Lys/Arg) consensus motif that occupies the P_2, $P_{1'}$, and $P_{2'}$ positions. Group III represents a novel substrate preference for MMP-9. As a whole, group III substrates were cleaved far less efficiently than either group I or group II. Nevertheless, in the context of the whole phage, at least one of the phages in this family (C11) was cleaved nearly as well as most of the group I substrates. An overriding feature of the group III substrates is the presence of Arg at P_2 and often at P_1. Although this recognition specificity is unexpected, it is corroborated by three other pieces of data. First, eight substrates from group I, including the four best peptides in this study, also have Arg at the P_2 position. Second, when peptide A11 from group I was mutated to contain Arg at both P_2 and P_1, the k_{cat}/K_m ratio for the peptide nearly tripled. Third, docking studies with a representative group III peptide illustrate the favorable interaction of Arg with the S_2 subsite (Fig. 1).

A walk across the model of the catalytic pocket of MMP-9, subsite by subsite, illustrates many of the features that are likely to guide substrate interactions, and confer recognition specificity. On the non-primed sides of the scissile bond, the structural features of the S3 and S2 subsites are consistent with the corresponding substrates that we selected. As with most MMPs, the S_3 subsite is a hydrophobic pocket that introduces a deviation from linearity into the active-site cleft. This deviation explains why Pro is favored at position P_3. At S_2, one finds the side chain of Asp-410 contributing its hydrophilic characteristics, and providing a potential salt bridge for the guanidinium group of Arg that is often found at P_2. MMP-13 also has an Asp residue at this position, and Arg occasionally occupies the P_2 positions of its substrates. In contrast, however, MMP-7 displays an Ala at this site rather than an Arg (3, 42), and none of the MMP-7 peptide substrates contain Arg at P_2 (32).

Moving across the cleft, one encounters the S_1 and $S_{1'}$ pockets. The S_1 space of MMP-9 is an extended hydrophobic surface that can accommodate a wide variety of amino acids (41). In fact, this is what we find. Like other MMPs, the $S_{1'}$ subsite is a deep hydrophobic pocket, and it is likely to be the most important substrate recognition point in the active site. This notion is supported by the fact that Leu and Ile residues dominate the $P_{1'}$ position. As discussed previously, the depth of this subsite also guides the binding of most of the small molecule antagonists of the MMPs (1, 3, 42).

The $S_{2'}$ subsite, a small hydrophobic surface, is the final significant recognition point for substrate in the cleft of MMP-9. The $S_{2'}$ subsite provides opportunity for flexible amino acids to adapt conformations that would allow interactions with the solvent on substrate binding at this position, including amino acids with small hydrophobic or large hydrophilic side chains. We find that Ser/Thr are preferred at this position. Even though two-thirds of the MMP-9 substrates derive their $P_{2'}$ position from an invariant Thr encoded by the vector, the remaining one-third derive Thr or Ser at their $P_{2'}$ positions by selection. Interestingly, the character of the $S_{2'}$ subsite differs among MMP-9, MMP-7, and MMP-13. In MMP-9 this subsite essentially comprises Asp-188, Gly-189, Leu-190, and Met-422 (D-G-L-M) (3, 42). The corresponding residues in MMP-7 are G-N-T-T, and in MMP-13 are S-G-L-I. These distinctions are consistent with the fact that the substrates for each protease also differ at $P_{2'}$ (see above).

Given this new understanding of substrate recognition by MMP-9, one can arrive at an optimal substrate consensus (Pro-Arg-(Ser/Thr)- ↓ -(Leu/Ile)-(Ser/Thr)). A comparison of this motif to previously reported substrates for MMP-9 reveals some interesting similarities and distinctions (Table III). For example, all of the collagen substrates and even aggrecan, tissue factor pathway inhibitor and galectin 3 contain a Pro at the P_3 position and a hydrophobic amino acid at the $P_{1'}$ position (6, 48, 50–52). However, none of the substrates contain residues found to be optimal at other positions. The remaining substrates have significantly lower homology to the MMP-9 consensus recognition motif of group I (8, 53).

These comparisons raise questions as to whether all of the physiologically relevant substrates for MMP-9 have been identified. We reasoned that a query of the protein data bases with the optimal recognition motif might reveal a short list of putative substrates. Indeed, this search revealed only nine human proteins that contain this sequence within their extracellular domain (Table IV). Although at this juncture these proteins can only be considered hypothetical substrates for MMP-9, many of them are functionally linked to MMP-9-related pathology. For example, MMP-9 influences skin blistering in the autoimmune disorder bullous pemphigoid (5, 55). Interestingly, two of the hypothetical substrates for MMP-9, ladinin 1 and desmoglein 3, are also associated with autoimmune skin blistering disorders (56, 57). Similarly, MMP-9 has been suggested as a regulator of the angiogenic switch in tumor development (11). Two of the hypothetical substrates for MMP-9, integrin β_5 and endoglin 1, are also involved in angiogenesis (49, 54, 58). Along with the presence of a potential MMP-9 cleavage site, these functional associations lead to important and testable hypotheses about physiologic substrates for MMP-9.

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